

IPF: Deep Scientific Journal Insights

Expert research briefing | Evidence through 2 September 2026

The field has entered a multi-mechanism treatment era, but not yet a precision-treatment era. The decisive advances are new phase 3 efficacy, human spatial biology, and increasingly credible risk stratification - while predictive biomarkers and proven fibrosis reversal remain absent.

Clinical	Nerandomilast approved; treprostinil phase 3 signal
Biology	Arrested alveolar repair in spatial fibrotic niches
Precision	Prognostic tools improving; no predictive biomarker
Research	Combination, sequencing, early disease, clinical events

Scope

Global peer-reviewed and regulatory evidence, weighted toward primary human studies and phase 3 trials. The report is designed for an advanced clinical-research audience and is not a patient-specific treatment recommendation.

Scope and direct answer

This briefing focuses on changes that materially alter how idiopathic pulmonary fibrosis (IPF) should be conceptualized, researched, and treated: phase 3 therapeutic evidence, human spatial and single-cell biology, genetics, prognostic biomarkers, quantitative imaging, and the strongest pipeline signals. It is not a complete clinical guideline and does not substitute for patient-specific prescribing decisions.

The central conclusion is that IPF has entered a multi-mechanism treatment era, but not yet a precision-treatment era. Nerandomilast is now approved in the United States and European Union, and two concordant phase 3 trials of inhaled treprostinil produced a large FVC-preservation signal. Yet no validated biomarker currently selects among approved agents, predicts treatment response, or establishes that an individual patient's fibrosis is reversible. Human spatial studies increasingly define IPF as an arrested-regeneration disease organized into pathologic epithelial-fibroblast-macrophage niches, rather than simply excess collagen deposition.

Executive insights

1. The therapeutic landscape changed in 2025-2026. Nerandomilast became the first new oral therapy approved for IPF in more than a decade. In FIBRONEER-IPF, adjusted mean FVC change at week 52 was -114.7 mL with 18 mg twice daily, -138.6 mL with 9 mg twice daily, and -183.5 mL with placebo. The treatment effect is real but modest, and the key composite of acute exacerbation, respiratory hospitalization, or death was not significantly improved.[1-3]
2. Treprostinil is the strongest late-stage investigational signal. TETON-2 showed median week-52 FVC changes of -49.9 versus -136.4 mL; TETON-1 showed -43.3 versus -196.2 mL. The pooled placebo-corrected effect was about 112 mL, with fewer clinical-worsening events and acute exacerbations. Cough, four-times-daily inhalation burden, and excess discontinuation are material limitations. As of the evidence cutoff, it was not yet approved specifically for IPF.[4,5]
3. FVC remains necessary but insufficient. It is reproducible, regulator-accepted, and prognostic at the group level, but it is not a direct measure of symptoms, fibrosis reversal, quality of life, or survival. Nerandomilast's positive FVC result without a significant composite-event benefit illustrates this separation. Future trials should preserve FVC while adding hierarchical clinical events, quantitative HRCT, patient-reported outcomes, and biomarker-defined enrichment.
4. IPF is increasingly an arrested alveolar-regeneration disorder. Human spatial transcriptomics identifies KRT5-negative/KRT17-positive aberrant basaloid cells, activated fibroblasts, and SPP1/APOE macrophages within TGF-beta, YAP/TEAD, and senescence-rich niches. The important contrast is with bleomycin injury: mice often complete epithelial repair and resolve fibrosis, while human IPF tissue remains trapped in maladaptive transitional states.[8-10]
5. Genetic heterogeneity is clinically meaningful before it is therapeutically actionable. MUC5B, telomere biology, rare pathogenic variants, and polygenic background define partially distinct liabilities. A 2026 genetic analysis combining more than 4,600 cases with more than 400,000 controls found that rare variants and very short telomeres identify a substantial minority, while common telomere and non-MUC5B polygenic risk adds independent information. These data support genetic counseling and telomere-aware management in selected patients, but not routine drug selection by polygenic score.[11]
6. MMP-7 and CCL18 are leading prognostic, not predictive, biomarkers. In 1,280 ISABELA participants, high baseline MMP-7 and CCL18 and early CCL18 change identified patients at higher risk of progression or death. They remain unqualified for routine treatment selection; assay harmonization, prespecified thresholds, prospective intervention studies, and regulatory qualification are missing.[12]
7. Quantitative CT is nearer to useful risk stratification than to autonomous diagnosis. Deep-learning fibrosis segmentation predicted progression and mortality in a prospective development cohort with external validation, and quantitative fibrosis burden can add information beyond visual labels. Scanner/protocol variation, opaque model behavior, dataset shift, and limited prospective workflow trials remain decisive barriers.[13]
8. Phase 2 success in IPF is unusually fragile. Pamrevlumab and zinpentraxin alfa failed phase 3 despite persuasive biology and earlier signals. In STARSCAPE, post hoc review showed that extreme placebo outliers had inflated the phase 2 zinpentraxin estimate. This argues for robust estimands, outlier-resistant sensitivity analyses, adequate duration, background-therapy stratification, and replication before biological narratives are treated as clinical truth.[6,7]

Therapeutic evidence: what the trials actually establish

Nerandomilast

Nerandomilast is a preferential PDE4B inhibitor with antifibrotic and immunomodulatory effects.

FIBRONEER-IPF randomized approximately 1,177 patients to 18 mg twice daily, 9 mg twice daily, or placebo, with background nintedanib, pirfenidone, or no antifibrotic permitted. The placebo-corrected week-52 FVC differences were 68.8 mL for 18 mg and 44.9 mL for 9 mg. Diarrhea occurred in 41.3%, 31.1%, and 16.0%, respectively.[1]

The high-value nuance is co-administration. Pirfenidone reduced nerandomilast exposure by about 50%; the 9 mg regimen was not effective with pirfenidone, which is why dose reduction to 9 mg is not appropriate in that combination. Diarrhea was most frequent with background nintedanib: in longer follow-up it occurred in 62.4% on nerandomilast 18 mg plus nintedanib, versus 31.2% with placebo plus nintedanib. Treatment discontinuation due to diarrhea reached 12.9% in the 18 mg plus nintedanib subgroup. These are subgroup observations and should not be mistaken for randomized head-to-head comparisons of combination strategies.[2]

Longer follow-up to approximately 76 weeks showed continued FVC-curve separation, but the composite clinical endpoint remained neutral. The mortality hazard ratio for 18 mg was 0.66 (95% CI 0.41-1.08), a potentially important but non-confirmatory signal.[2]

Regulatory status is now clear: FDA approval for adult IPF occurred on 7 October 2025; EU-wide marketing authorization was granted on 15 July 2026. The EU recommended dose is 18 mg twice daily.[3]

Interpretation: nerandomilast is a validated FVC-sparing therapy. Evidence does not yet prove symptom improvement, fibrosis reversal, or survival benefit. In practice and research, its greatest unresolved question is sequencing versus combination with nintedanib or pirfenidone.

Inhaled treprostinil

The TETON program produced unusually concordant phase 3 efficacy. TETON-2 reduced median FVC loss by 86.5 mL and clinical worsening from 39.0% to 27.2% (HR 0.71). TETON-1 produced a 130.1 mL placebo-corrected FVC difference and HR 0.67 for clinical worsening. Pooled analyses suggested a 31% reduction in clinical worsening and 48% reduction in acute exacerbation, without a statistically significant survival benefit.[4,5]

The signal is biologically plausible: prostacyclin receptor signaling raises intracellular cAMP and may inhibit fibroblast proliferation and extracellular-matrix production, not merely dilate pulmonary vessels. But tolerability and treatment burden matter. In TETON-2 cough occurred in 48.3% versus 24.1%, and discontinuation occurred in 33.6% versus 24.7%. Four daily nebulized sessions create a real-world adherence test that an efficacy trial cannot fully resolve.[4]

Interpretation: if confirmed through regulatory review, treprostinil would support multi-pathway therapy and make monotherapy-versus-combination trials urgent. Its effects should not be extrapolated to imply that all pulmonary vasodilators are antifibrotic.

Established antifibrotics

Pirfenidone and nintedanib remain proven FVC-sparing therapies. Registry analyses using causal-inference methods suggest a survival advantage for antifibrotic treatment, but residual confounding remains possible: in IPF-PRO, the mortality HR was 0.53 with a 95% CI of 0.28-1.03.[14] No convincing evidence establishes superiority of one over the other for survival. Selection remains driven by comorbidity, interaction profile, tolerability, lifestyle, access, and patient preference.

Disease biology: a spatial ecosystem, not a linear cascade

The most coherent modern model begins with susceptible alveolar epithelium exposed to repeated injury. Ageing, telomere dysfunction, MUC5B-related mucociliary biology, environmental exposures, ER stress, mitochondrial dysfunction, and senescence reduce regenerative capacity. AT2 cells enter transitional or aberrant basaloid states rather than completing AT1 differentiation. These cells occupy the same spatial niches as CTHRC1-like activated fibroblasts, myofibroblasts, and SPP1/APOE macrophages.[8-10]

Three implications follow:

- The target is a niche. Blocking one soluble mediator may fail because epithelial state, matrix stiffness, macrophage signals, and fibroblast mechanotransduction reinforce one another.
- Timing probably matters. Advanced honeycomb lung contains fixed architectural loss; an intervention that restores epithelial differentiation may work best at the leading edge of fibrosis or in preclinical interstitial lung abnormalities.
- Animal translation must be mechanism-specific. Single-dose bleomycin is valuable for acute injury pathways but poorly models the persistent human failure of alveolar regeneration. Candidate selection should require human-cell, human-tissue, and spatial target-engagement evidence.

Genetics and telomere biology

MUC5B rs35705950 is the strongest common susceptibility signal, but susceptibility is not prognosis and the variant does not select therapy. Rare variants in TERT, TERC, RTEL1, PARN and other telomere-maintenance genes, and very short telomeres even without an identifiable rare variant, define a clinically important subgroup. Telomere disease can coexist with marrow, liver, skin, or premature-greying phenotypes and can influence immunosuppression tolerance and transplant planning.

The 2026 integrative analysis supports a layered model: rare high-impact variants, MUC5B, an IPF polygenic score, and a telomere-length polygenic score contribute partly independent information. This is promising for risk architecture, but clinical use requires ancestry-diverse validation and proof that testing changes outcomes.[11]

Biomarkers and imaging

The field should separate four biomarker roles:

Role	Current best candidates	Readiness
Diagnostic support	HRCT/UIP pattern, MUC5B genotype, epithelial proteins	HRCT is established; blood/genetic panels are adjunctive
Prognosis	FVC/DLCO trajectory, GAP, fibrosis extent, MMP-7, CCL18	Physiologic and imaging measures are usable; proteins remain investigational
Treatment response	Serial FVC, possibly quantitative HRCT and CCL18	No validated predictive biomarker
Target engagement	Drug-specific plasma/tissue markers	Useful in trials; not routine care

MMP-7 and CCL18 are reproducible risk markers, but their incremental value over age, sex, FVC, DLCO, oxygen use, and HRCT must be shown in decision-impact studies.[12] Quantitative CT can measure fibrosis burden, vessel volume, and texture more continuously than categorical UIP labels. The 2024 deep-learning segmentation study is an important prospective/external-validation advance, but clinical deployment still requires locked algorithms, protocol standardization, calibration monitoring, and evidence that using the score improves decisions.[13]

Why promising IPF trials fail

Pamrevlumab showed no significant week-48 FVC benefit in ZEPHYRUS-1: -260 versus -330 mL, difference 70 mL (95% CI -60 to 190; P=0.29), with no secondary or patient-reported benefit.[6] STARScape found no benefit of zinpentraxin alfa and stopped for futility; extreme placebo declines had materially distorted the prior phase 2 estimate.[7]

These failures do not prove that CTGF or pentraxin biology is irrelevant. They show that target plausibility, biomarker movement, and small-trial FVC separation are not interchangeable with replicated clinical efficacy. Better development standards include:

1. Replicated phase 2 evidence or an adaptive seamless design.
2. Robust estimands with prespecified handling of death, transplant, dropout, and extreme FVC values.
3. Stratification by background therapy and explicit interaction testing.
4. Human spatial target validation and pharmacodynamic confirmation.
5. Endpoints that link FVC to symptoms, quantitative structure, exacerbations, hospitalization, and survival.

Pipeline signals ranked by evidentiary maturity

High priority:

- Admilparant (LPA1 antagonist): phase 2 at 60 mg twice daily reduced 26-week ppFVC decline in IPF, with a treatment difference of 1.4 percentage points (95% CI -0.1 to 3.0). The IPF confidence interval crossed zero, while the PPF cohort was positive. ALOFT-IPF phase 3 is the appropriate test; primary completion was estimated for October 2026.[16,17]
- Deupirfenidone: phase 2b ELEVATE-IPF reported a 91 mL 26-week difference versus placebo at 825 mg three times daily. The active pirfenidone comparator and Bayesian design are strengths; phase 3 replication and full peer-reviewed comparison remain necessary.[18]

Promising but early:

- Rentosertib (TNIK inhibitor): in a 71-patient, 12-week Chinese phase 2a trial, 60 mg once daily produced a mean FVC increase of 98.4 mL versus a 20.3 mL decline with placebo. This is hypothesis-generating: short duration, small dose groups, geographic concentration, and efficacy as a secondary/exploratory signal make phase 3 confirmation essential. A China phase 3 trial began in 2026.[19]
- Taladegib (Hedgehog inhibitor): early phase data suggested improvements in FVC and quantitative CT measures; a global dose-ranging phase 2b trial is the key test.[20]

Cautionary signal:

- Bexotegast (alpha-v-beta-6/alpha-v-beta-1 integrin inhibitor): BEACON-IPF was discontinued after an independent safety review because of an imbalance in unadjudicated IPF-related adverse events despite an early efficacy signal. This is a reminder that target engagement and FVC trends cannot override an uncertain benefit-risk profile.[21]

Most important research conclusions

1. The near-term clinical research question is no longer "does any drug slow FVC decline?" It is "which combination, sequence, and dose preserves clinically meaningful function with acceptable burden?"
2. The most plausible precision axis is an integrated endotype combining telomere/genetic liability, epithelial-regeneration state, quantitative HRCT trajectory, and circulating matrix/immune markers.
3. The field needs trials in earlier disease and interstitial lung abnormalities, but only with careful overdiagnosis safeguards and endpoints that prove net benefit.
4. Reversal should require convergent evidence: sustained FVC improvement, declining quantitative fibrosis burden, improved gas transfer or exercise physiology, and a mechanistically coherent biomarker shift.
5. Acute exacerbation remains the major uncontrolled event. The negative EXAFIP cyclophosphamide result and weak steroid evidence argue for prospective platform trials rather than therapeutic tradition.[15]

Evidence boundaries

- Trial populations underrepresent very advanced disease, major comorbidity, frailty, women, and some ancestral groups.
- FVC effects cannot be directly compared across trials because background therapy, estimands, missing-data methods, populations, and follow-up differ.
- Subgroup and pooled mortality signals are exploratory unless prespecified and multiplicity-controlled.

- Spatial-omics studies are mechanistically powerful but often use small numbers of transplant-stage lungs and cannot alone establish causality.
- Regulatory approval establishes a favorable population-level benefit-risk assessment; it does not identify the optimal sequence or combination for an individual patient.

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